Object Segmentation Based on Watershed and Graph Cut

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***Abstract*—In this paper we select the inner contour and outer contour of the target object, and then apply watershed algorithm to divide the region between the inner contour and the outer contour into series of smaller regions. Each smaller region is regarded as a node, and the small regions contains of the inner contour are regarded as the source s, and the small regions contains of the outer contour are regarded as the sink t. Then from the source to the sink establishes a graph. Merging the capacities of parallel edges, deleting unnecessarily edges, multi-source and multi-sink can be converted to the single source and the single sink. And then minimum energy can be obtained by the maximal-flow cut, and the target object can be extracted. The experiment proves that this algorithm is effective in the segmentation than the other traditional algorithms, and has better performance in the running time.**

Keywords- graph cut; watershed algorithm; max flow/min cut; image processing

I. INTRODUCTION

Image segmentation is important in the computer vision, and simultaneously it is also the foundation of pattern recognition and image analysis. Many experts have proposed various algorithms in image segmentation. However, there has no one algorithm which can be used in every kinds of image, there are always some limitations. There are many algorithms based on region: region growing, split-and-merge and others, but they are often weak or blurry in the boundaries places. The algorithms based on active contour such as snakes, deformable templates and the shortest path, cannot be easily generalized to the three dimensional image. In [1] the algorithm using graph cut based on the boundary and the region was presentted, and energy function was established. The target object can be obtained by max flow/min cut algorithm. Graph cut could be effective to extract object boundary, and plays more and more important role in the computer vision. In [2] the watershed algorithm was proposed, but it has the over-segmentation phenomenon. The object extracted by the watershed always has little meanings, generally there should merge regions. But the watershed algorithm locates boundaries well, and can maintain small differences inside each small region [3].

In this paper, we choose the watershed algorithm to pre-segment the image, and then segment the image basing on the small regions instead of the pixels. So the number of nodes and edges for the graph cut algorithm is reduced very much, and it improves the speed significantly. At the same time, because the object regions are selected largely possible and back regions are small possible, the running speed of the algorithm can be raised. This paper also simplified multi-sources multi-sink graph to s-t graph, which also can reduce the running speed times. The algorithm also overcame the problems in [4] to obtain the inner contour and outer contour through the morphology dilation, but there had the problems that the inner contour or the outer contour may not exist, which would influence the segmentation.

II. RELATED THEORY

A. Related Graph Theory

Segmentation can be converted into the problem of finding the minimum of the energy function. Graph cut is the algorithm to find the global optimization solution using min cut in the graph theory. Using graph cut to segment the image contains several steps as follows:

1) Construct graph structure. Convert the image into an undirected graph or directed graph, and each pixel is defined as a node. Intensities and gradient and other criteria between neighboring pixels are transformed as the edges costs in the graph. There are two additional terminal nodes: the source s and the sink t. A 2D image can be converted a directed graph as shown in Fig.1(a).

2) Construct the energy function. Graph cut contains the information between boundaries and regions, and these can be converted into the problems to find the labels when energy function had the minimum. The energy function equation can be expressed as (1).

$$E = \sum_{\{p,q\}} V_{p,q} + \sum_{p \in P} D_p \tag{1}$$

The term $V_{p,q}$ is the penalty for a discontinuity between p and q, and it may be based on local intensity gradient. D_p is

the data energy, and it measures how well labels fits pixel given the observed data.

3) Using max flow/min cut to find the global optimization solution of the energy function. The minimum of the energy function is the accurate segmentation of the image. The result segmentation as shown in Fig.1(b). In section *C*, we will introduce the max flow/min cut algorithm used in this paper.

B. Multi-source Multi-sink

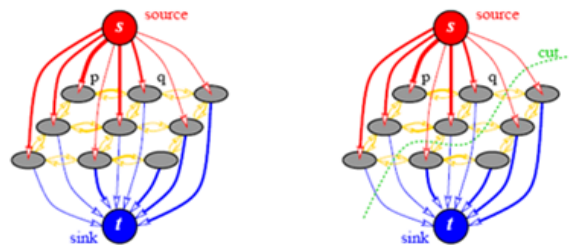
The image can be regarded as a directed graph $G=(V,\varepsilon)$, where V are the vertices which are the pixels in the image, and ε are defined as a set of directed edges that connect these nodes. For the multi-source multi-sink graph, we adopts the method in [4] that the multiple source and the multiple sink can be converted to the single source and the single sink, merging the capacity of the parallel edges, deleting unnecessarily edges. That is to say, $V_s=\{v_{s1},v_{s2},.....,v_{sn}\}$ is the multiple source which can be converted as a single new node V_1, and $V_t=\{v_{t1},v_{t2},.....,v_{tn}\}$ is the multiple sink which can be converted a single new node V_2. Each node v_i which has be adjacent to source, the cost of the edge form V_1 to the node v_i is the sum cost of the node in V_s which are connected to v_i. In the same way, the cost of edge from v_i which has adjacent to the sink to V_2 can be obtained. Then the graph $G=(V,\varepsilon)$ can be converted into the new graph $G'=(V',\varepsilon')$, where $V'=V-V_s-V_t+V_1+V_2$. The graph $G=(V,\varepsilon)$ and the graph $G'=(V',\varepsilon')$ have the same energy, and these are proved in [4]. Multi-source multi-sink as shown in Fig. 2, can be converted to s-t graph as shown in Fig. 3.

C. Max Flow/Min Cut

Max flow/min cut algorithm can be used to find the minimum energy or approximation energy for network. It is very popular in computer vision. A network $G=(V,\varepsilon)$, where V is the vertices, and ε is defined as a set of directed edges, source s, sink t, and flow f. The general structure of the algorithm is: from s and t to find the search tree SS and TT, we will find augmenting path [5].

1) "growth" stage
2) "augmentation" stage
3) "adoption" stage
4) If the augmenting path can not be found in step 1), the min cut would be found, and exit the algorithm. Otherwise, iteratively repeat 1)-3).

Using the algorithm above, we will find the augmenting path from source s to sink t, as shown in Fig.4. In the Fig.4 thick line is the path from source to sink, and the dotted line denotes that the pixel connects to source or sink, and the dotted sphere denotes that the pixel neither connects to source nor sink. In [6] the author added topology constraints for these pixels to overcome holes problem.



(a) A Graph *G* (b) A Cut on *G*

Figure 1. Graph and Cut in [1]

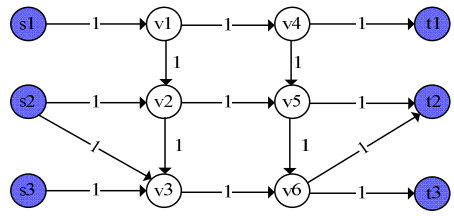


Figure 2. Multi-source Multi-sink Graph

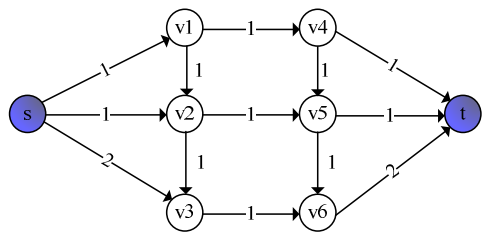


Figure 3. s-t Graph

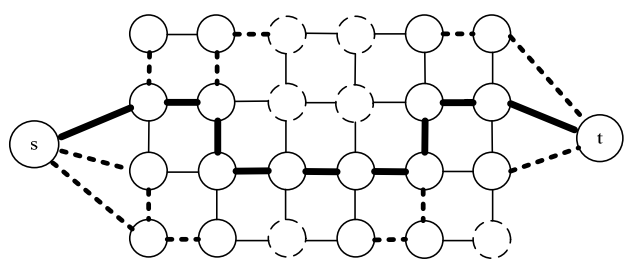


Figure 4. A Path from s to t

D. Watershed Algorithm

The watershed algorithm came from geomorphology. Vincent and Soille presented an efficient algorithm based on immersion simulation in 1991, and the algorithm is the most applied at the present time. Regard the image as natural landform covered by water, and gradients of the pixels in the image can be regarded as the altitude of the pixels. Pierce a hole at the minimum of the regions, and then irrigate water slowly to the surface. When water will immerse ground gradually, catchment basins can be formed. We will build a dam at where the water from different catchment basins will be joined. At the end of the immersion process, every catchment basin will be surrounded by the dams. The boundaries of the

catchment basins, which are the dams, are the watersheds of the image.

The algorithm includes the sorting step and the flooding step, and it is described in detail and provided in a pseudo C language in [2].

III. THE ALGORITHM BASED ON GRAPH CUT AND WATERSHED

The boundaries of the image can be divided two parts: object s and background t.

1) Select a region as the inner contour IC in the target object with the paint brush.
2) Select a region as the outer contour OC out of the target object with the paint brush.
3) Use the watershed algorithm to pre-segment the region from the inner contour to the outer contour, then the region becomes series of small regions.
4) Regard the region with the inner contour as source s, and regard the region with the outer contour as the sink t. Then convert the graph to single source and sink, to form the new graph *G*.
5) Using the max flow/min cut to find the minimum of the energy function.

Steps 1) to 5) of the algorithm are shown in Fig. 5-Fig.11. Using the watershed algorithm to pre-segment the region, and labels for every small region is supposed as Fig. 6. The last segmentation is shown in Fig. 11.

A. Pre-segment Image and Establish Graph

After watershed algorithm to pre-segment image between the inner contour and outer contour, the new graph *G* is established regarding the small regions as the nodes.

First establish two-dimension array *Label(length)(width)*=3, length is the length of the image, and width is the width of the image. The array can be used to record the label of each pixel at initial state in the image. Then select the inner contour of the target object with the paint brush, and record it position, *Label(i, j)*=0, there *(i, j)* is the position of a point at inner contour. In the same way, select the outer contour of the target object with the paint brush, and record it position, *Label(i, j)*=1, there *(i, j)* is the position of any one at outer contour. The region between the inside contour and the outside contour *Label(i,j)*=2, will be segmented. So $Label(i,j) \in \{0,1,2,3\}$.

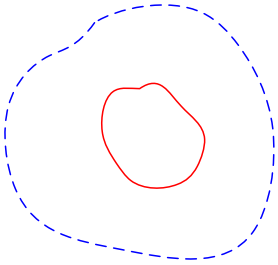


Figure 5. Inner Contour and Outer Contour (real line is inner contour, and broken line is outer contour)

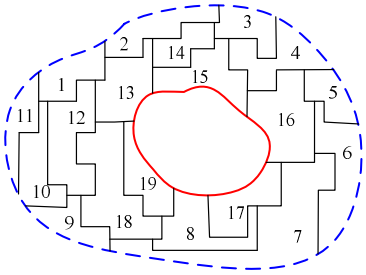


Figure 6. Pre-segmentation for the Region of the Inner Contour and the Outer Contour

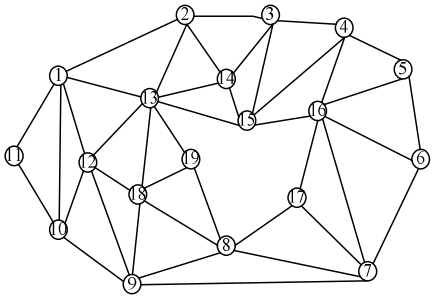


Figure 7. Graph for the Pre-segmentation Region

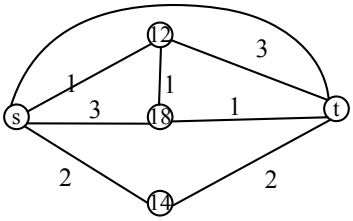


Figure 8. Single Source and Single Sink Graph

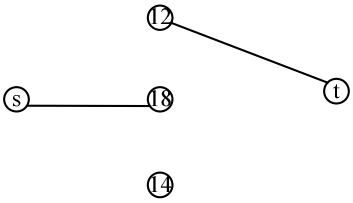


Figure 9. Min Cut Result

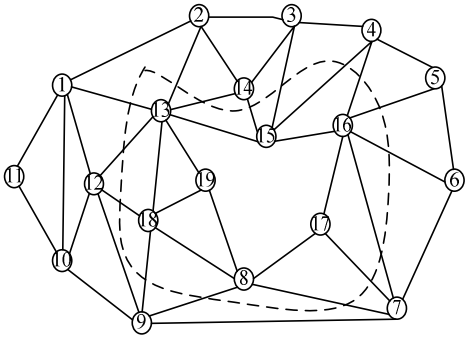


Figure 10. Min Cut in the Graph(broken line is the cut)

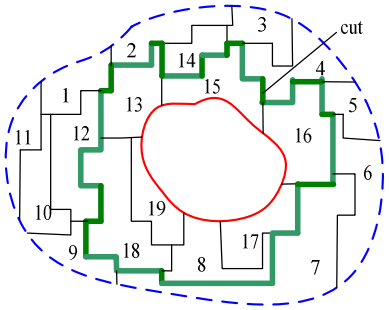


Figure 11. Min Cut in the Real Image

Using the watershed algorithm to pre-segment the image for Label*(i, j)*=2 region, then the image will divide into *n* small regions. Each small region will have different marks *f*[*k*], *k*=1…*n*, and then establish graph structure for these small region, which is also the process that seeking for the adjacency regions of each small region. The steps of the algorithm as follows:

1) *k*=1;

2) Traversal all pixels in this region labeled '*k*' and seek for all the pixels' adjacency pixels. If there is the adjacency pixel which is label '*x*', then the region labeled '*k*' is adjacency to the region labeled '*x*'. After traversing the region labeled '*k*', all adjacency regions for the region labeled '*k*' will be found. If there is a adjacency pixel which has *Label(i,j)*=0 in the region labeled '*k*', where *i* and *j* is the pixel's horizontal ordinate and vertical ordinate, the region k is a source. If there is a adjacency pixel which has *Label(i,j)*=1 in the region labeled '*k*', where *i* and *j* is the pixel's horizontal ordinate and vertical ordinate, then the region *k* is a sink.

3) *k*=*k*+1, if *k*>*n*, the algorithm ended, otherwise return to the step 2) and continue to seek for the next region's adjacency region.

After the region between the inner contour and outer contour is pre-segment, each small region' adjacency regions will be found according to above algorithm. Then the new graph *G* can be established relying on the adjacency information.

B. Edge Weight

An adjacency relation of pixels in the image can be represented by 4-neighborhood or 8-neighborhood. In order to avoid possible ambiguity, the paper represent the image as 8-neighborhood. The cost of the edge for the adjacency pixels represents the pixels similarity. The paper uses the formula in [4] as follows:

$$C(i,j) = (g(i,j) + g(j,i))^6 \qquad (2)$$

Where $g(i,j) = \exp(-grad_{ij}(i) / \max_k(grad_{ij}(k)))$. $grad_{ij}(k)$ is the gradient magnitude of image pixel at location *k* from *i* to *j*. $\max_k(grad_{ij}(k))$ is the maximum of the gradient magnitude. In the pre-segmentation regions, $grad_{ij}(i)$ is the average value of the gradient magnitudes of the region labeled *i*. $C(i,j)$ represents the cost of the adjacency regions *i* and *j*. The watershed algorithm provides a good super set of object boundaries, and this approximation produces reasonable results and improves the speed significantly [3].

IV. EXPERIMENT

The algorithm in this paper is implemented by C++ and Matlab, and all the results are obtained at the same computer which configuration is 2G memory, Intel(R) Core(TM) 2 Duo processor and XP operating system. Through the algorithm to segment series of images, and the algorithm in this paper compare the segmentation results and the running time to the algorithm in [4]. Fig. 12 is the results using the algorithm in the paper, and Fig. 13 is the results using the algorithm in [4], where left side is the original image, and right side is the segmentation. From Fig.12 and Fig. 13, we can see that the segmentation results are better than the results in [4]. At the same time, the selection for the inner contour and outer contour can reflect the results of segmentation. Reference [4] relies more on the selecting contour than our paper method.

Table 1 is the comparison of the running time between the algorithms in the paper and in [4]. From the table 1, we can see that the operating efficiency of the paper algorithm go far beyond the algorithm in [4].

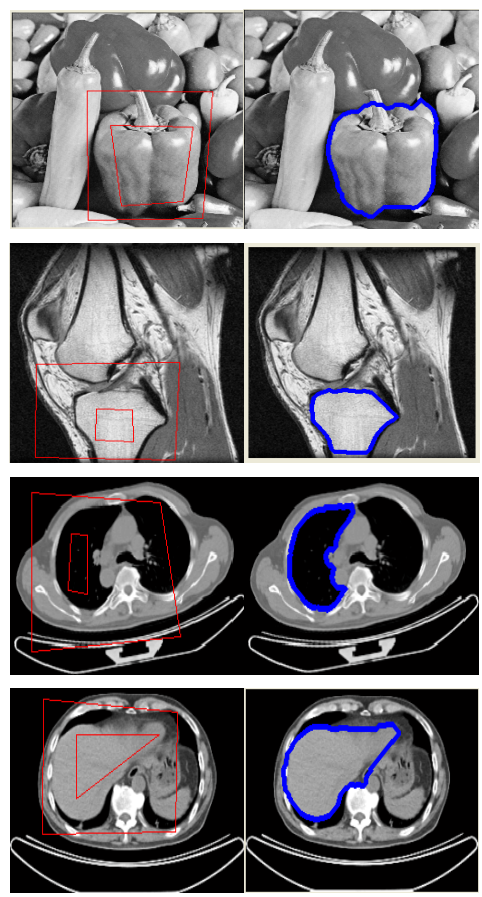


Figure 12. the Segmentation Results in the Paper

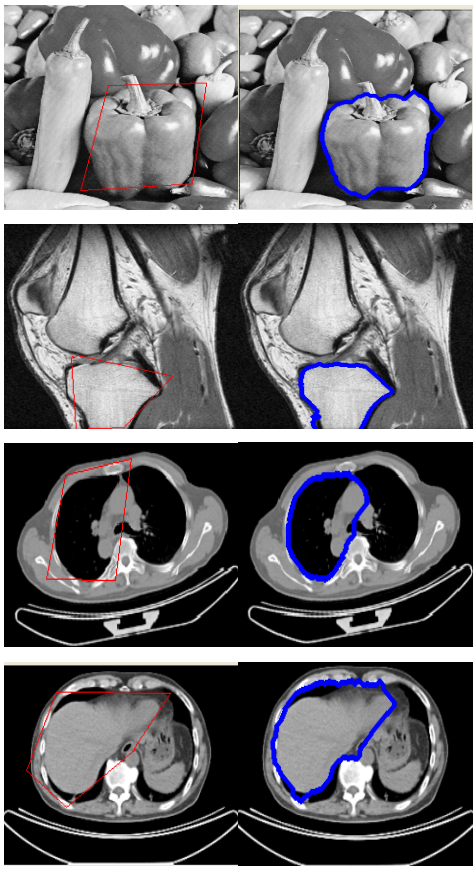


Figure 13. the Segmentation Results in [4]

TABLE I. THE RUNNING TIME

image	The algorithm in the paper(s)	The algorithm in [4] (s)
peppers256*256	0.6250	2.4375
Knee400*400	0.4375	1.4382
Lung249*162	0.4063	1.2173
Liver297*202	0.4844	1.8204

V. CONCLUSIONS

The paper combines the watershed with graph cut. Firstly, select inner contour and outer contour, then pre-segment the regions between the inner contour and the outer contour. Finally, we use the max flow algorithm to find the minimum energy. The algorithm has the better segmentation results, and improves the computing efficiency. Because the paper still uses the methods that combining the regions and boundaries, the object contour can be obtained accurately. In the computing efficiency aspect, the paper uses several ways to improve it as follows: selecting the inner contour and outer contour reduces the searching regions, and regard the pre-segmentation small region as a pixel, and convert multi-source multi-sink to single source and single sink.

ACKNOWLEDGMENT

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